Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

1. Let m be the Lebesgue measure on $[0,1]$ and $I_A(x)$ denote the indicator function of A. Consider the following sequence of random variables on $(\Omega = [0,1], \mathcal{F} = \mathcal{B}[0,1], P = m)$: For $n \geq 1$, $X_n(\omega) = n^2 I_{(\frac{1}{n+1}, \frac{1}{n})}(\omega)$ and $X(\omega) = 0$ for $\omega \in \Omega$.
   (i) Show $X_n \to X$ almost surely (P).
   (ii) Does $X_n \to X$ in probability ?
   (iii) Show $X_n \to X$ in $L^p \equiv L^p(\Omega, \mathcal{F}, P)$ for $0 < p < 1$, but this convergence does not hold for $1 \leq p < \infty$.

   (7+7+(7+7) = **28** points)

2. Let X, Y be two **independent** random variables on some probability space $(\Omega, \mathcal{F}, P)$, with marginal laws P_X, P_Y on $(I\!R, \mathcal{B}(I\!R))$. Recall that X and Y independent iff their joint law is the *product measure* of their marginal laws, i.e $P_{(X,Y)} = (P_X) \times (P_Y)$ on $(I\!R^2, \mathcal{B}(I\!R^2))$. Let m is the Lebesgue measure on $(I\!R, \mathcal{B}(I\!R))$ and $m^2 \doteq m \times m$ is the Lebesgue measure on $(I\!R^2, \mathcal{B}(I\!R^2))$. Let f, g be two non-negative functions in $L^1(I\!R, \mathcal{B}(I\!R), m)$.

   - (a) If X and Y are independent and $E(X^2) < \infty$, $E(Y^2) < \infty$, then carefully prove
   $$Var(X+Y) = Var(X) + Var(Y).$$
   - (b) Suppose $P_X \ll m, P_Y \ll m$, with $\frac{dP_X}{dm} = f$ and $\frac{dP_Y}{dm} = g$. Then prove
   $$P_{(X,Y)} \ll m^2, \quad \frac{dP_{(X,Y)}}{dm^2} = f.g \quad a.e\ (m^2)$$
   - (c) If f is continuous and bounded on $I\!R$, so is $f * g$.
   - (d) If $\lambda \doteq P_X * P_Y$ Prove $\frac{d\lambda}{dm} = f * g$ a.e (m), f and g are as in (b).
   - (e) If Z is a random variable with $P_Z = \lambda$ as in (d) above, then how is Z related to X and Y ? What does the result in (d) say about the law of Z ?

   (8+8+8+8+(4+4)= **40** points)

3. (i) Show that a $Binomial(n = 10, p = 0.1)$ random variable exists:
   In other words, using steps of the Caratheodory extension theorem, exhibit a probability space $(\Omega, \mathcal{F}, P)$ and a random variable $X : \Omega \to I\!R$, such that its p.m.f. is:
   $$P(X = k) = f(k) = \binom{10}{k}(0.1)^k(0.9)^{10-k}, \ k \in \{0, 1, \ldots, 10\},$$
   and $P(X = k) = f(k) = 0$, otherwise.

(ii) Compute the following Lebesgue integrals (give a numerical answer)

$$E(X) = \int_{\Omega} X(\omega) dP(\omega), \quad Var(X) = \int_{\Omega} (X(\omega) - E(X))^2 \, dP(\omega).$$

You can state & use any pre-measure theory facts you know (if needed), but justify all the measure theory steps.

(iii) For each $n \geq 1$, let μ^n be a probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$. Under what conditions can we say that there is a probability measure P on $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$ such that $P(A \times \mathbb{R}^\infty) = \mu^n(A)$ for any $A \in \mathcal{B}(\mathbb{R}^n)$, for any $n \geq 1$? Describe a sequence of random variables $\{X_n : n \geq 1\}$ defined on $(\Omega = \mathbb{R}^\infty, \mathcal{F} = \mathcal{B}(\mathbb{R}^\infty), P)$ such that the joint distribution of $(X_1, \ldots, X_n)$ is given by μ^n, for all $n \geq 1$.

(iv) Using part (iii) or otherwise, show that there exists $\{X_n : n \geq 1\}$ which is an i.i.d. sequence of $Binomial(n = 10, p = 0.1)$ random variables.

(v) Using any of the problems above (or otherwise), show that if

$$Y_n = \frac{1}{n}(X_1 + \ldots + X_n)$$

then $\{Y_n : n \geq 1\}$ converges in probability to 1.

(4×7 +9= **37** points)